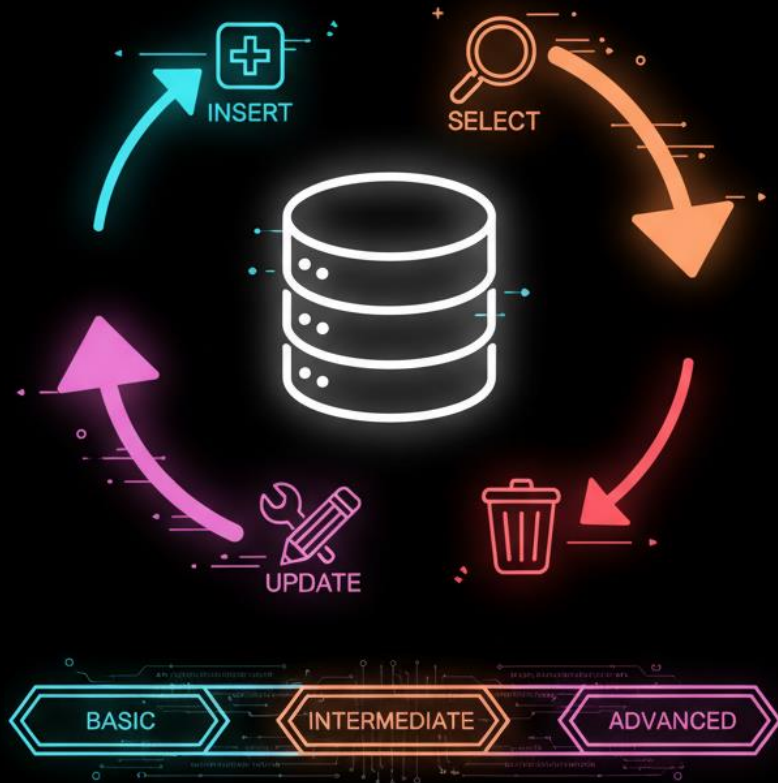


Foundations & Basics

SQL Statements & Execution Order



By: Zherou A. Majied

SQL Statement Structure: Writing Order

When writing an SQL statement, clauses are written in this sequence. Only SELECT and FROM are mandatory for a basic query.

Step	Clause	Purpose
1	SELECT	Choose columns or expressions to return
2	DISTINCT	Remove duplicate rows
3	TOP or TOP PERCENT	Limit the number of rows returned (cannot be used with OFFSET)
4	FROM	Specify the table(s) or data source
5	JOIN	Combine data from multiple tables
6	WHERE	Filter rows before grouping
7	GROUP BY	Group rows for aggregation
8	HAVING	Filter groups after aggregation
9	ORDER BY	Sort the result set
10	OFFSET ... FETCH	Paginate results (cannot be used with TOP)

Note: A SELECT can exist without a FROM, e.g. SELECT @@VERSION

SQL Statement Execution: Actual Processing Order

The database engine executes clauses in a different order than how they are written.

Step	Execution Phase	Explanation
1	FROM & JOIN	Identify tables and combine data
2	WHERE	Filter rows before grouping
3	GROUP BY	Group rows for aggregation
4	HAVING	Filter aggregated groups
5	SELECT & DISTINCT	Return final columns, remove duplicates
6	ORDER BY & TOP	Sort and limit the result set

